Diffusion-Based 3D Coronary Centerline Reconstruction from Sparse-View X-Ray Angiography

Project Summary

Coronary angiography provides interventional cardiologists with only 2D X-ray projections during live procedures, yet clinical decisions such as whether a stenosis requires stenting depend on 3D vessel geometry and hemodynamic quantities (e.g., wall shear stress) that 2D images cannot reveal directly, a gap responsible for the estimated 40-50% of angiographically ambiguous lesions where visual assessment alone is unreliable. Reconstructing 3D coronary geometry from sparse, non-simultaneous 2D projections is a fundamentally ill-posed inverse problem, since multiple 3D structures can project identically onto the same images, and classical geometric methods fail on the low-contrast, overlapping, thin-vessel anatomy typical of angiography. This two-phase project addresses that gap: Phase 1 learns the conditional distribution of plausible 3D vessel centerlines given their 2D projections via a conditional diffusion model, following recent precedent (DeepCA, AortaDiff, 2025) that generative models trained on anatomical priors can resolve this ambiguity where classical structure-from-motion cannot; Phase 2 then predicts hemodynamic fields directly from the reconstructed centerline (following an anisotropic RBF-decoder approach), toward the longer-term goal of real-time, wire-free hemodynamic guidance during coronary interventions.

Approach

Phase 1 reconstructs the 3D coronary centerline from 2 sparse, non-simultaneous 2D X-ray projections, generated as digitally reconstructed radiographs via TIGRE (Biguri et al., 2016) from ImageCAS CTA volumes, with an independent rigid perturbation applied to the second view to simulate cardiac motion between shots (following DeepCA's protocol, Wang et al., WACV 2025). Ground-truth centerlines and radius profiles are extracted directly from ImageCAS's expert segmentation masks via skeletonization, with each point additionally labeled by local topology (endpoint, regular, or bifurcation) to support the fixed-topology diffusion process below. The reconstruction model is a conditional diffusion architecture operating natively on the centerline representation, a 1D-UNet denoiser over ordered (x, y, z, radius) points, conditioned on both projections and their calibrated projection matrices via cross-attention so the model can reason about epipolar correspondence between views, following AortaDiff's (2025) centerline-diffusion design, which outperformed generic point-cloud diffusion baselines (PVD, DiT-3D) on an analogous vascular reconstruction task. A classical, non-learned baseline (epipolar-constraint point matching and triangulation) is implemented alongside the diffusion model to establish a reconstruction-quality floor. Evaluation uses Chamfer L2 distance and a threshold-based overlap metric (DeepCA's protocol) on held-out cases, with particular attention to high-foreshortening and vessel-overlap subsets where reconstruction is hardest. Phase 2 will build on the resulting centerlines to predict hemodynamic fields (pressure/FFR, wall shear stress, velocity) directly, via an anisotropic RBF-decoder approach operating on the same centerline representation.

Related Work/ References:

Phase 1(3D centerline reconstruction) :

1- "TIGRE: a MATLAB-GPU toolbox for CBCT image reconstruction." Biomedical Physics & Engineering Express, 2016.
(dataset preparation, evaluation metrics)

2- "AortaDiff: Volume-Guided Conditional Diffusion Models for Multi-Branch Aortic Surface Generation."
(centerline-diffusion architecture, primary precedent)

3- "3DGR-CAR: Coronary Artery Reconstruction from Ultra-sparse 2D X-Ray Views with a 3D Gaussians Representation."
(sparse-view dataset construction precedent, ImageCAS split logic)

4- "Diffusion Probabilistic Models for 3D Point Cloud Generation".
(point-cloud diffusion baseline)

5- "DiT-3D: Exploring Plain Diffusion Transformers for 3D Shape Generation."
(point-cloud diffusion baseline)

Data & Tools :

1- "ImageCAS: A large-scale dataset and benchmark for coronary artery segmentation."
Computerized Medical Imaging and Graphics, 2023.

2- "TIGRE: a MATLAB-GPU toolbox for CBCT image reconstruction." Biomedical Physics & Engineering Express, 2016.

3- "Anatomy-Grounded Synthetic Coronary Angiography for Geometry-Informed Multi-View Matching." arXiv:2606.28474, 2026.

4- "Patient-Specific Dynamic Digital-Physical Twin for Coronary Intervention Training: An Integrated Mixed Reality Approach." arXiv:2505.10902, 2025.

Phase 2, hemodynamics from centerline (planned):

1- "From Centerlines to Hemodynamics: Anisotropic RBF Decoders for Coronary Arteries." 2026.
(target Phase 2 architecture)

Clinical motivation & background:

1- "Advances in the computational assessment of disturbed coronary flow and wall shear stress: a contemporary review."

2- Hidden fluid mechanics: Learning velocity and pressure fields from flow visualizations."

(validates the underlying physical premise; per-patient optimization limits clinical deployment, motivates the generalizable, trained-once approach here)

3- "A novel physics-based model for fast computation of blood flow in coronary arteries" (FAST).

4- "Time-resolved blood flow measurement in interventional angiography using 4D DSA."

Dataset:

Training data is derived from ImageCAS, a public benchmark of 1000 coronary CT angiography (CCTA) volumes with expert-annotated vessel segmentation masks. Since neither raw CT volumes nor paired 2D-3D angiography data exist at scale, data preparation follows a 3-step pipeline:

1- 3D ground-truth centerlines and radius profiles are extracted directly from each case's segmentation mask via classical skeletonization.

2- synthetic 2D X-ray projections are generated from each CTA volume using TIGRE (Biguri et al., 2016), which computes a direct Beer-Lambert attenuation line integral from Hounsfield-derived density with no intermediate material-classification step. This was chosen over a material-decomposition-based simulator after the latter was found to absorb contrast-enhanced vessel voxels into a generic tissue class, weakening vessel-to-background contrast; TIGRE avoids this failure mode and matches the toolchain used by our primary reference pipeline (DeepCA, Wang et al., WACV 2025). Two non-simultaneous views are generated per case, with an independent rigid perturbation applied to the second view to approximate cardiac motion, following DeepCA's motion-simulation protocol.

3- cases are split at the case level (never by view, to avoid leakage) into train/validation/test sets, following the training-set scale used on ImageCAS by prior sparse-view reconstruction work (3DGR-CAR, MICCAI 2024). Prior to training, centerlines and projections undergo preprocessing: centerline coordinates are converted to physical units and normalized, variable-length centerlines are padded to a fixed size with an accompanying mask, and projection images are intensity-normalized — all normalization statistics computed from the training split only, to avoid information leakage into validation/test. This yields paired (2D projection, 3D centerline ground truth) training examples without requiring any real intraoperative angiography data.

Evaluation Plan:

1- Reconstruction quality is assessed using two point-set metrics on held-out validation and test cases, both following DeepCA's (Wang et al., WACV 2025) evaluation protocol:

- **Chamfer L2 distance** between predicted and ground-truth centerlines.
- **Threshold-based overlap metric Ot(d)** (evaluated at multiple distance thresholds), particularly informative under motion/deformation, where exact point correspondence isn't expected.

2- To contextualize these numbers rather than report model performance in isolation, a classical baseline is evaluated alongside the diffusion model:

- **Epipolar-constraint matching + triangulation**: standard multi-view geometry (not a specific paper's method), used to establish a reconstruction-quality floor.

3- Evaluation additionally targets the hardest, most clinically relevant conditions rather than only average-case results:

- **Stress-test subset** of cases with high foreshortening angle or vessel overlap, failure modes repeatedly flagged in the sparse-view reconstruction literature (DeepCA, Wang et al., WACV 2025; 3DGR-CAR, Fu et al., MICCAI 2024), with subset construction our own.

4- For clinical interpretability beyond aggregate point-cloud accuracy:

- **Tube-surface visualization**: reconstructed centerlines rendered via a deterministic geometric sweep on the predicted radius profile, alongside ground truth, including visible-stenosis cases, grounded in how minimal lumen diameter is computed clinically via Quantitative Coronary Angiography (not a paper-sourced method).

Risks & Scope

The primary risk is a sim-to-real gap: all training data is synthetic, generated via TIGER projections from CTA volumes, with no real intraoperative angiography used for training, a limitation DeepCA and 3DGR-CAR both flag explicitly, and one this project only partially addresses through a conditional fine-tuning step (Step 3.3) contingent on real ICA data becoming available, not a guaranteed deliverable. A second risk concerns training stability: hyperparameters are seeded from AortaDiff's reported setup, but that paper trained on only 18 cases with 3D-volume conditioning, versus roughly 960 cases with 2D-projection conditioning here, so batch size, learning rate, and convergence behavior will likely require empirical retuning, adding uncertainty to the training timeline. The project also depends on external computers, since DRR generation and full-scale model training require a CUDA GPU (Google Colab) unavailable on the local development machine (Apple Silicon M4), introducing a workflow dependency on cloud-compute access and its limits. Architecturally, the model assumes centerline topology is fixed and correctly extracted, diffusing only node position and radius, so reconstruction quality is not evaluated on cases where branch topology itself is ambiguous. Finally, in terms of scope, this proposal covers Phase 1 (3D centerline reconstruction) only; Phase 2 (hemodynamic field prediction from the reconstructed centerline) is a planned follow-on and falls outside this deliverable's evaluation and timeline.

Ethical Implications

This project uses ImageCAS, a public, de-identified CT angiography dataset, with no real patient data collected or used directly during Phase 1 development, and produces no clinical deployment or diagnostic output at this stage, reconstructions are evaluated against ground truth in a research setting only. Even so, the eventual clinical motivation raises considerations worth naming now. First, a reconstruction model trained entirely on synthetic DRR projections may not generalize reliably to real intraoperative imaging, and any future clinical use would require rigorous validation on real angiography data and human-in-the-loop oversight rather than autonomous decision-making, a wrong 3D reconstruction feeding into a stenting decision carries real patient risk. Second, ImageCAS's demographic and scanner-specific composition may not represent the full range of patient anatomy and imaging equipment encountered clinically, and this should be examined before any claim of generalizability. Finally, should this work progress toward Phase 2 and eventual clinical translation, it would require institutional review, regulatory clearance, and prospective clinical validation, none of which are in scope for this academic project, this proposal represents a feasibility study of the underlying reconstruction problem, not a clinical tool.